

```
Processando config 29, seed 0, teste: BlockAlphabit
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L$  = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.9802323e-08, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 2
Value of the statistic : 0.96
p-value of test : 0.62
```

```
-----
CPU time used : 00:00:00.03
```

Generator state:

16777216 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells =  $2^L$  = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 5.88
p-value of test : 0.66

CPU time used : 00:00:00.02

Generator state:

33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation

Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 151.76
p-value of test : 0.07

CPU time used : 00:00:00.03

Generator state:

50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32603.36
p-value of test : 0.74

CPU time used : 00:00:00.03

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 155.26
p-value of test : 0.88

CPU time used : 00:00:00.03

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 283.76
p-value of test : 0.80

CPU time used : 00:00:00.02

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -1.23
p-value of test : 0.89

CPU time used : 00:00:00.02

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	35.93
p-value of test	:	0.29

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	22.38
p-value of test	:	0.87

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	26.20
p-value of test	:	0.75

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	23.43
p-value of test	:	0.55

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	13.65
p-value of test	:	0.55

CPU time used : 00:00:00.06

Generator state:

134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 1

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	50.20
p-value of test	:	0.86

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	64.04
p-value of test	:	0.40

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	151.16
p-value of test	:	0.68

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	56.05
p-value of test	:	0.51

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	27.76
p-value of test	:	0.68

CPU time used : 00:00:00.06

Generator state:
150994944 bits have been read.

===== Summary results of Alphabit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216
Number of statistics:	17
Total CPU time:	00:00:00.33

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

```
For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
```

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 0.37
p-value of test                    : 0.83
```

```
-----
CPU time used                      : 00:00:00.03
```

```
Generator state:
```

```
167772160 bits have been read.
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 2
```

```
smultin_MultinomialBitsOver test:
```

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
```

Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	8
Value of the statistic	:	10.31
p-value of test	:	0.24

CPU time used : 00:00:00.02

Generator state:

184549376 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216	
Number of cells = 2^L =	256
Expected number per cell =	65536
Hashing =	FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	128
Value of the statistic	:	115.40
p-value of test	:	0.78

CPU time used : 00:00:00.00

Generator state:

201326592 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

smultin_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 0.00097653271, Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom : 32768
Value of the statistic : 32679.70
p-value of test : 0.63
```

```
-----
CPU time used : 00:00:00.03
```

Generator state:

218103808 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2
```

sstring_HammingIndep test:

```
-----
N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0
```

Counters with expected numbers >= 10

```
-----
Number of degrees of freedom : 177
Chi-square statistic : 164.29
p-value of test : 0.74
```

```
-----
CPU time used : 00:00:00.02
```

Generator state:

234881024 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	302.34
p-value of test	:	0.53

CPU time used	:	00:00:00.02
---------------	---	-------------

Generator state:

251658240 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic	:	-0.36
p-value of test	:	0.64

CPU time used	:	00:00:00.00
---------------	---	-------------

Generator state:

268435456 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	47.86
p-value of test	:	0.04

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	41.58
p-value of test	:	0.10

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	21.96
p-value of test	:	0.91

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	28.76
p-value of test	:	0.27

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	6.79
p-value of test	:	0.96

CPU time used : 00:00:00.06

Generator state:

285212672 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 2

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	60.65
p-value of test	:	0.52

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	82.47
p-value of test	:	0.04

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	168.99
p-value of test	:	0.30

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	50.30
p-value of test	:	0.72

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	32.68
p-value of test	:	0.43

CPU time used : 00:00:00.06

Generator state:

301989888 bits have been read.

===== Summary results of Alhabit =====

Version:	TestU01 1.2.3
Generator:	ufire_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216
Number of statistics:	17
Total CPU time:	00:00:00.28

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alphabit:  nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1
```

```
-----
Test Results for Delta = 1.0000
```

```
Number of degrees of freedom      : 2
Value of the statistic             : 1.07
p-value of test                   : 0.59
```

```
-----
CPU time used                     : 00:00:00.03
```

Generator state:

318767104 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 4
```

smultin_MultinomialBitsOver test:

```
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
```

Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 7.85
p-value of test : 0.45

CPU time used : 00:00:00.00

Generator state:
335544320 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For $\Delta > -1$, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 114.49
p-value of test : 0.80

CPU time used : 00:00:00.03

Generator state:
352321536 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32892.13
p-value of test : 0.31

CPU time used : 00:00:00.03

Generator state:

369098752 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 196.65
p-value of test : 0.15

CPU time used : 00:00:00.02

Generator state:

385875968 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom	:	305
Chi-square statistic	:	299.58
p-value of test	:	0.58

CPU time used : 00:00:00.00

Generator state:

402653184 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 1.68
p-value of test : 0.05

CPU time used : 00:00:00.02

Generator state:

419430400 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	29.40
p-value of test	:	0.60

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	25.40
p-value of test	:	0.75

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	40.54
p-value of test	:	0.14

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	26.26
p-value of test	:	0.39

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	10.65
p-value of test	:	0.78

CPU time used : 00:00:00.06

Generator state:

436207616 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 4

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	55.66
p-value of test	:	0.70

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	66.35
p-value of test	:	0.33

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	189.70
p-value of test	:	0.05

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	62.51
p-value of test	:	0.29

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	26.33
p-value of test	:	0.75

CPU time used : 00:00:00.05

Generator state:

452984576 bits have been read.

===== Summary results of Alphabit =====

Version:	TestU01 1.2.3
Generator:	ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits:	16777216

Number of statistics: 17
Total CPU time: 00:00:00.27

All tests were passed

```
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
Starting Alhabit: nb = 16777216
Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
```

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	5.24
p-value of test	:	0.07

CPU time used : 00:00:00.02

Generator state:

469761792 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic : 0.84
p-value of test : 0.9991 *****
```

CPU time used : 00:00:00.02

Generator state:

486539008 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

```
Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 3.784895e-06, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 128
Value of the statistic : 128.72
p-value of test : 0.47
```

CPU time used : 00:00:00.02

Generator state:

503316224 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32256.92
p-value of test : 0.98

CPU time used : 00:00:00.03

Generator state:

520093440 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 176.77
p-value of test : 0.49

CPU time used : 00:00:00.02

Generator state:
536870656 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 8
```

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 312.24
p-value of test : 0.38

CPU time used : 00:00:00.00

Generator state:
553647872 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 8
```

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : -0.16
p-value of test : 0.57

CPU time used : 00:00:00.02

Generator state:
570425088 bits have been read.

```
*****
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	26.22
p-value of test	:	0.75

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	27.45
p-value of test	:	0.65

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	27.99
p-value of test	:	0.67

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	18.80
p-value of test	:	0.81

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	9.55
p-value of test	:	0.85

CPU time used : 00:00:00.06

Generator state:

587202304 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 8
```

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	55.17
p-value of test	:	0.72

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	60.32
p-value of test	:	0.54

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	164.90
p-value of test	:	0.38

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	62.66
p-value of test	:	0.28

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	33.05
p-value of test	:	0.42

CPU time used : 00:00:00.06

Generator state:
603979264 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3

```

Generator:      ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25
The following tests gave p-values outside [0.001, 0.9990]:
(eps means a value < 1.0e-300):
(eps1 means a value < 1.0e-15):

```

Test	p-value
-----	-----
2 MultinomialBitsOver, L = 4	0.9991
-----	-----
All other tests were passed	

```

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
      Starting Alhabit:  nb = 16777216
      Version: TestU01 1.2.3
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

```

```

*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin:  /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen:  r = 0,  s = 32,  w = 16

```

```

smultin_MultinomialBitsOver test:
-----
N = 1,  n = 16777216,  r = 0,  s = 32,  L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:
Delta = 1,  Mu = 2.9802323e-08,  Sigma = 1

```

```

-----
Test Results for Delta = 1.0000

Number of degrees of freedom      : 2
Value of the statistic            : 3.85
p-value of test                   : 0.15

-----
CPU time used                     : 00:00:00.02

```

```

Generator state:
620756480 bits have been read.

```

```

*****
HOST = matheus-IdeaPad-3-15ALC6, Linux

```

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16
```

smultin_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 2.0861626e-07, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 8
Value of the statistic : 15.29
p-value of test : 0.05
```

CPU time used : 00:00:00.00

Generator state:

637533696 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

```
ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16
```

smultin_MultinomialBitsOver test:

```
-----
N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE
```

```
Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE
```

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

```
Delta = 1, Mu = 3.784895e-06, Sigma = 1
```

Test Results for Delta = 1.0000

```
Number of degrees of freedom : 128
Value of the statistic : 112.29
p-value of test : 0.84
```

CPU time used : 00:00:00.02

Generator state:
654310912 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32654.30
p-value of test : 0.67

CPU time used : 00:00:00.03

Generator state:
671088128 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 177  
Chi-square statistic              : 145.41  
p-value of test                   : 0.96  
-----
```

```
-----  
CPU time used                     : 00:00:00.02  
-----
```

Generator state:
687865344 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingIndep test:

```
-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0  
-----
```

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 305  
Chi-square statistic              : 261.28  
p-value of test                   : 0.97  
-----
```

```
-----  
CPU time used                     : 00:00:00.00  
-----
```

Generator state:
704642560 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

sstring_HammingCorr test:

```
-----  
N = 1, n = 524288, r = 0, s = 32, L = 32  
-----
```

```
-----  
Normal statistic                  : 0.73  
p-value of test                   : 0.23  
-----
```

```
-----  
CPU time used                     : 00:00:00.02  
-----
```

Generator state:
721419776 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	19.50
p-value of test	:	0.96

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	39.57
p-value of test	:	0.14

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	43.54
p-value of test	:	0.08

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	20.24
p-value of test	:	0.73

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	14.41
p-value of test	:	0.50

CPU time used : 00:00:00.06

Generator state:
738196992 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 16

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	51.85
p-value of test	:	0.82

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	62.46
p-value of test	:	0.46

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	157.25
p-value of test	:	0.55

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	43.61
p-value of test	:	0.90

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	38.69
p-value of test	:	0.19

CPU time used : 00:00:00.06

Generator state:

754973952 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25

All tests were passed

XXX
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
XXX

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 4
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 2
Value of the statistic : 0.53
p-value of test : 0.77

CPU time used : 00:00:00.02

Generator state:
771751168 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 16
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 7.58
p-value of test : 0.48

CPU time used : 00:00:00.02

Generator state:

788528384 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128

Value of the statistic : 128.71
p-value of test : 0.47

CPU time used : 00:00:00.00

Generator state:
805305600 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 33043.55
p-value of test : 0.14

CPU time used : 00:00:00.03

Generator state:
822082816 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 177  
Chi-square statistic              : 201.31  
p-value of test                   : 0.10  
-----
```

```
-----  
CPU time used                     : 00:00:00.00  
-----
```

Generator state:

838860032 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingIndep test:

```
-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0  
-----
```

Counters with expected numbers ≥ 10

```
-----  
Number of degrees of freedom      : 305  
Chi-square statistic              : 327.61  
p-value of test                   : 0.18  
-----
```

```
-----  
CPU time used                     : 00:00:00.02  
-----
```

Generator state:

855637248 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

sstring_HammingCorr test:

```
-----  
N = 1, n = 524288, r = 0, s = 32, L = 32  
-----
```

```
-----  
Normal statistic                  : -0.32  
p-value of test                   : 0.63  
-----
```

CPU time used : 00:00:00.02

Generator state:
872414464 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin
unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom	:	32
ChiSquare statistic	:	52.68
p-value of test	:	0.01

Test on the values of the Statistic M

Number of degrees of freedom	:	31
ChiSquare statistic	:	37.72
p-value of test	:	0.19

Test on the values of the Statistic J

Number of degrees of freedom	:	32
ChiSquare statistic	:	47.52
p-value of test	:	0.04

Test on the values of the Statistic R

Number of degrees of freedom	:	25
ChiSquare statistic	:	28.80
p-value of test	:	0.27

Test on the values of the Statistic C

Number of degrees of freedom	:	15
ChiSquare statistic	:	14.58
p-value of test	:	0.48

CPU time used : 00:00:00.06

Generator state:

889191680 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config29_seed0.bin

unif01_CreateBitBlockGen: r = 0, s = 32, w = 32

swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom	:	62
ChiSquare statistic	:	44.86
p-value of test	:	0.95

Test on the values of the Statistic M

Number of degrees of freedom	:	62
ChiSquare statistic	:	57.97
p-value of test	:	0.62

Test on the values of the Statistic J

Number of degrees of freedom	:	160
ChiSquare statistic	:	146.61
p-value of test	:	0.77

Test on the values of the Statistic R

Number of degrees of freedom	:	57
ChiSquare statistic	:	56.99
p-value of test	:	0.48

Test on the values of the Statistic C

Number of degrees of freedom	:	32
ChiSquare statistic	:	35.26
p-value of test	:	0.32

CPU time used : 00:00:00.05

Generator state:
905968640 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin, unif01_CreateBitBlockGen:
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.25

All tests were passed